



MediKiosk: AI-Powered Digital Clinical Intake Platform

An accessible, multilingual self-service kiosk empowering OPD patients through adaptive voice triage, document OCR intelligence, and instant clinical synthesis for physicians.

💡 Proposed Solution & End-to-End Workflow



✔ Core Innovation & Clinical Methodology

- ✔ **Adaptive Conversational Triage:** Context-aware Gemini 1.5 engine asks targeted, 1-by-1 clinical questions (onset, severity, aggravating factors) rather than static forms.
- ✔ **Dual-Mode Inclusivity:** Elderly and rural patients can either speak naturally in Hindi/English or tap large, clear emoji touch buttons.
- ✔ **Document Intelligence:** Multi-engine OCR digitizes handwritten prescriptions and lab reports, stitching them into an automatic timeline.



Technical Approach & System Flow

Frontend (Kiosk & Physician Dashboard)

Next.js, Tailwind CSS, Framer Motion for responsive audio waveforms, Web Speech API fallback.

Backend & Microservices

FastAPI asynchronous service layer handling audio buffers, real-time WebSockets, and state validation.

AI & Document Core

Google Gemini 1.5 Pro (Clinical Triage & SOAP synthesis), Cloud Vision OCR, Google Cloud Speech-to-Text & TTS (Indic en-IN, hi-IN).

Data, Privacy & ABDM Integration

Firebase Firestore session tracking, FHIR-compliant schema mapping, explicit DPDP Act 2023 patient consent logs.

Microservices Data Pipeline

Patient Kiosk (Voice / Touch Input)



FastAPI Async Orchestrator & Session Manager



Speech & Gemini AI
(Conversational Triage)

Cloud Vision OCR
(Prescription Digitization)



Real-Time Doctor Dashboard (SOAP Summary & ABDM)



Feasibility, Challenges & Mitigation

Domain	Operational Challenge	MediKiosk Mitigation Strategy
Audio Environment	High ambient noise levels in crowded Indian government hospital waiting areas.	Integration of directional noise-filtering ASR, complemented by automatic tactile on-screen tap buttons for every question.
Clinical Safety	Risk of AI hallucination or misleading automated patient self-diagnoses.	AI strictly acts as an intake reporter without diagnostic autonomy. Generates physician-editable draft SOAP summaries only.
Emergency Triage	Critical patients waiting in kiosk queue with silent acute myocardial infarction or stroke.	Real-time Red-Flag NLP detector triggers instant priority flashing alerts on nursing triage consoles.
Elderly Accessibility	Hesitation and lack of digital literacy among rural and senior citizen patients.	Zero-navigation UI with auto-spoken local audio instructions (Hindi/Regional) guiding each step intuitively.
Ayurvedic OPD	Standard intake tools lack holistic Ayurvedic diagnostic parameters.	Integrated Ayush module collecting Dashavidha Pariksha, Ahara-Vihara, and Agni/Koshtha indicators.

Impact, Clinical Value & Key Benefits



For Physicians & Healthcare Staff

Eliminates the "blank-slate" problem. Doctors receive a structured, chronological digest of chief complaints, past medications, and lab values before the patient sits down.

For Rural & Low-Literacy Patients

Empowers patients to communicate freely in their native dialect without feeling rushed, ensuring no critical symptom or allergy is forgotten.

Systemic & National ABDM Alignment

Standardizes paper-based OPD records into FHIR-ready digital health data right at the entry point of the healthcare delivery pipeline.

80%

Reduction in Documentation Friction

Cuts initial history-taking review down from 4 minutes to under 45 seconds.

Research, References & Regulatory Alignment



Clinical Evidence

BMJ Open (2017): Global study highlighting that average primary care consultations in India last approximately 2 minutes, creating acute diagnostic compression and incomplete history capture.



ABDM & FHIR Protocols

National Health Authority (NHA): Conforms strictly to Ayushman Bharat Digital Mission (ABHA) guidelines and HL7 FHIR standards for seamless hospital information exchange.



DPDP Act 2023 Governance

Data Privacy & Consent: Incorporates explicit, multi-lingual audio and touch consent before processing sensitive patient health data, with isolated per-session document storage.

Smart India Hackathon 2026 Submission

Bakhtiyarpur College of Engineering | Team MediKiosk | PS ID: 2047

✓ Standard 6-Slide Compliance Verified

Image Sources



<https://www.dmaarogya.com/wp-content/uploads/2026/08/Kiosk-Machine-blog.webp>

Source: www.dmaarogya.com



<https://www.altexsoft.com/static/blog-post-featured/2023/10/c39f55f9-c7f0-4aa1-841a-86f2c1c1eb02.jpg>

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